OCA OCPP 2.0.1 / 2.1 PLUGFEST & CONFERENCE AUSTRALIA

21–23 October 2026 · Canberra

From grid signal to bidirectional charging profile

open, standard, end to end.

Home energy management that coordinates CSIP-AUS dynamic operating envelopes and OpenADR 3.1 price signals into OCPP 2.1 bidirectional charging profiles the charging station validates, responding quickly to price and network events, including 1-minute-notice Minimum System Load and Lack of Reserve events.

1 Two signals in, one OCPP 2.1 profile out

every component open source

Network operator CSIP-AUS server
ANU BSGIP Cactus v1.2 test harness

→

Open Dynamic Export (ODE)
CSIP-AUS client dynamic envelopes

→

Home Assistant EEMS
Green · OpenADR VEN bridge + CitrineOS-HA

→

CitrineOS CSMS
OCPP 2.1 / 2.0.1 / 1.6 open-source CSMS

→

EVerest → Plugfest stations
charging stations validate profiles

→

Electric vehicle V2G
ISO 15118-20 EV charge + · discharge −

SEP2 · IEEE 2030.5 CSIP-AUS

MQTT limits

REST CitrineOS API

OCPP 2.1 SetChargingProfile

ISO 15118-20 DC_BPT

OpenADR 3.1 VTN
price server
AU tariff events · 5/30-min

OpenADR 3.1 PRICE · EXPORT_PRICE

OCPP 2.1 SetChargingProfile · TxProfile · CentralSetpoint · setpoint / limit / dischargeLimit in W

Prices travel with the power: absolutePriceSchedule on the profile · CostUpdated (California Pricing) · SetDefaultTariff. Not reliant on the OCPP 2.1 DER control block.

2 Proven on 17 October 2026

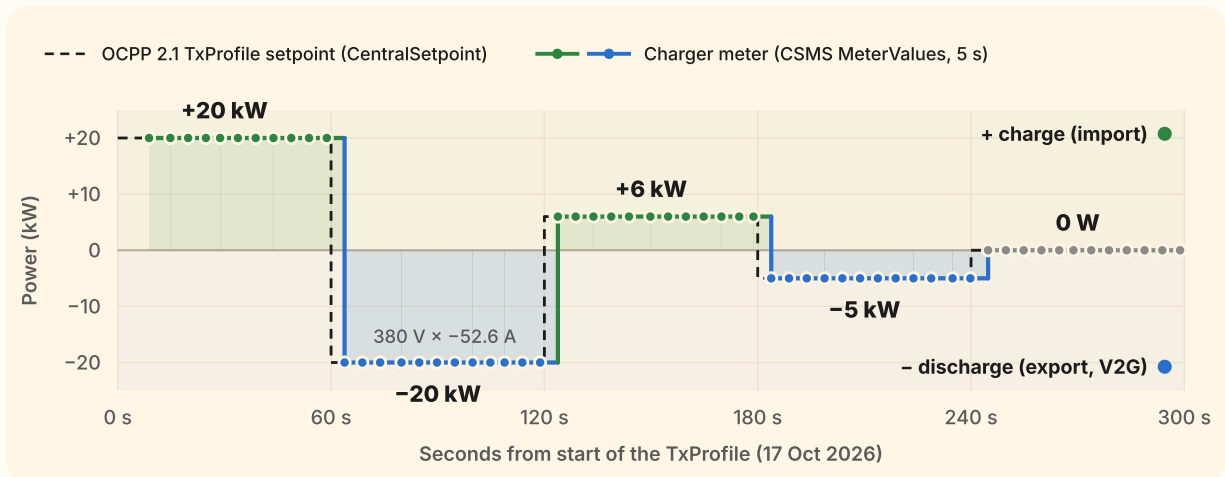
ANU BSGIP Cactus CSIP-AUS v1.2 test harness · EVerest DC V2G simulator

16/16	16/16	±20 kW	±25 kW	600 s	100%
LOA-11 import limits	**LOA-11 import limits**	**tracked exactly**	**DC V2G rating**	**schedule-ahead**	**open source**
AC charger · run 74858	DC V2G charger · run 74865	OCPP 2.1 CentralSetpoint	ISO 15118-20 DC_BPT	≤10 OCPP profile periods	every component

Schedule-ahead: the bridge sends the next 600 s of CSIP-AUS controls as OCPP 2.1 profile periods, so each limit takes effect at its start time.
In run 74865 every decrease applied 0.5–0.6 s early from the held profile, including 0 W.

3 One TxProfile, five setpoints

EVerest DC V2G simulator · ISO 15118-20 DC_BPT · ±25 kW rated · 60 s per step



4 What the demo shows

live, on the Plugfest floor

- **Export-limit clip** during V2G discharge
- **Import-limit clip** during charging
- **Load-limit clip** site load envelope
- **1-minute MSL / LOR** short-notice events
- **Local congestion** network constraint
- **Price events** OpenADR 3.1 PRICE

5 Draft AS 5438: three protocols an EEMS must support

SEP2
via CSIP-AUS
IEEE 2030.5 · AS 5385:2023
Primary path

OCPP 2.1
bidirectional EV charging
SetChargingProfile · TxProfile
Primary path

SunSpec Modbus 1.2
inverters
hybrid inverter
Secondary path

AS 5438 function-map timing targets

≤5 s	≤30 s	≤15 s
read / write	active-power limit	comms-loss fail-safe

Mark Purcell AM
purcell.id.au Pty Ltd · Sunshine Coast, QLD

Also at the Plugfest as a CSMS under test: CitrineOS on Home Assistant Green, Security Profile 1

github.com/purcell-lab/ocpp-plugfest

OCA OCPP Plugfest & Conference · Canberra · 21–23 Oct 2026